

2025.4.27 Yokohama

Brain Cache: Generative AI as a Cognitive Exoskeleton for Externalizing, Structuring, and Activating Knowledge

Long Ling, Intelligent Big Data Visualization Lab, Tongji University, China



CHI 2025

April 26–May 1, 2025 in Yokohama, Japan

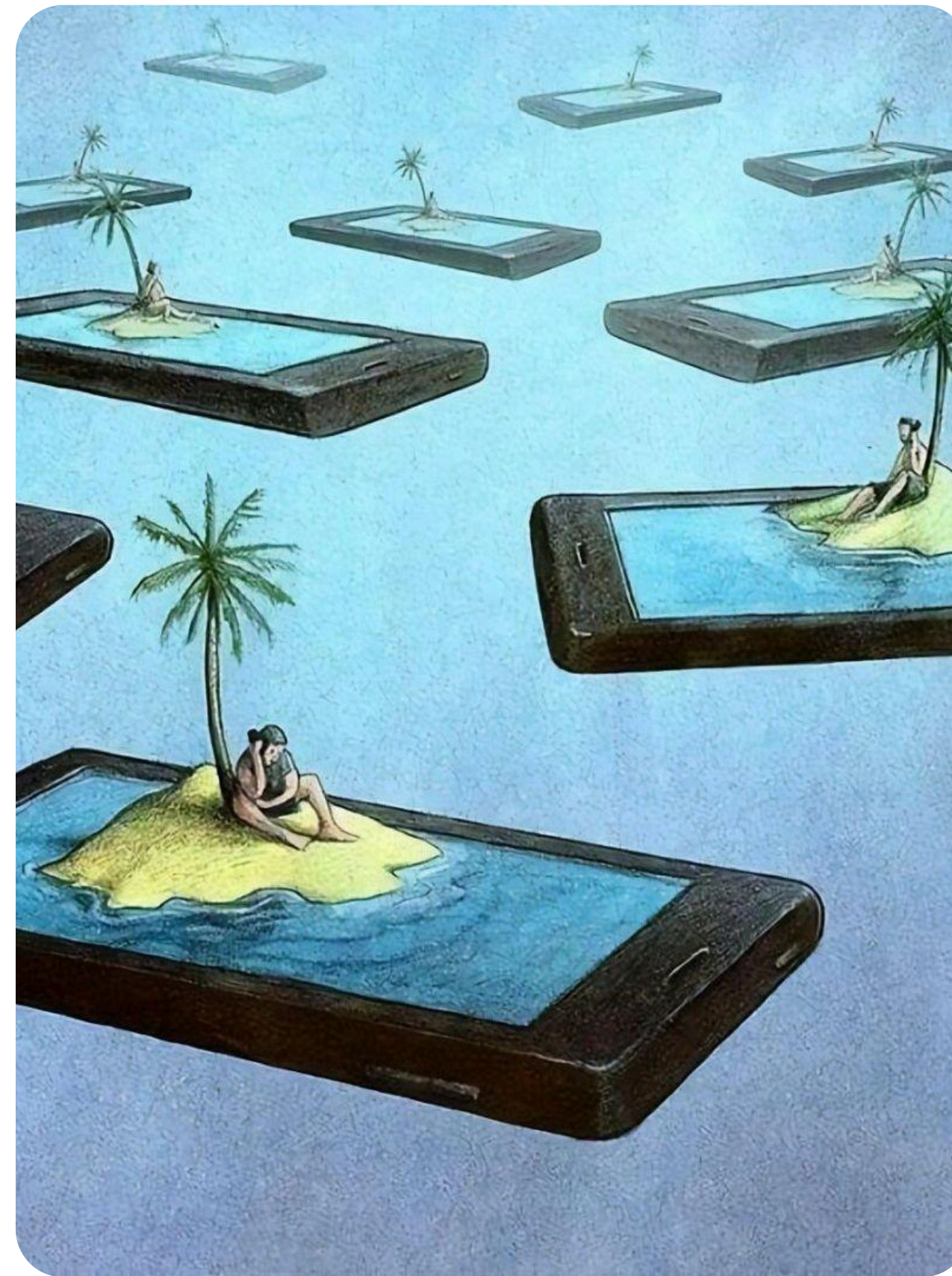


Background

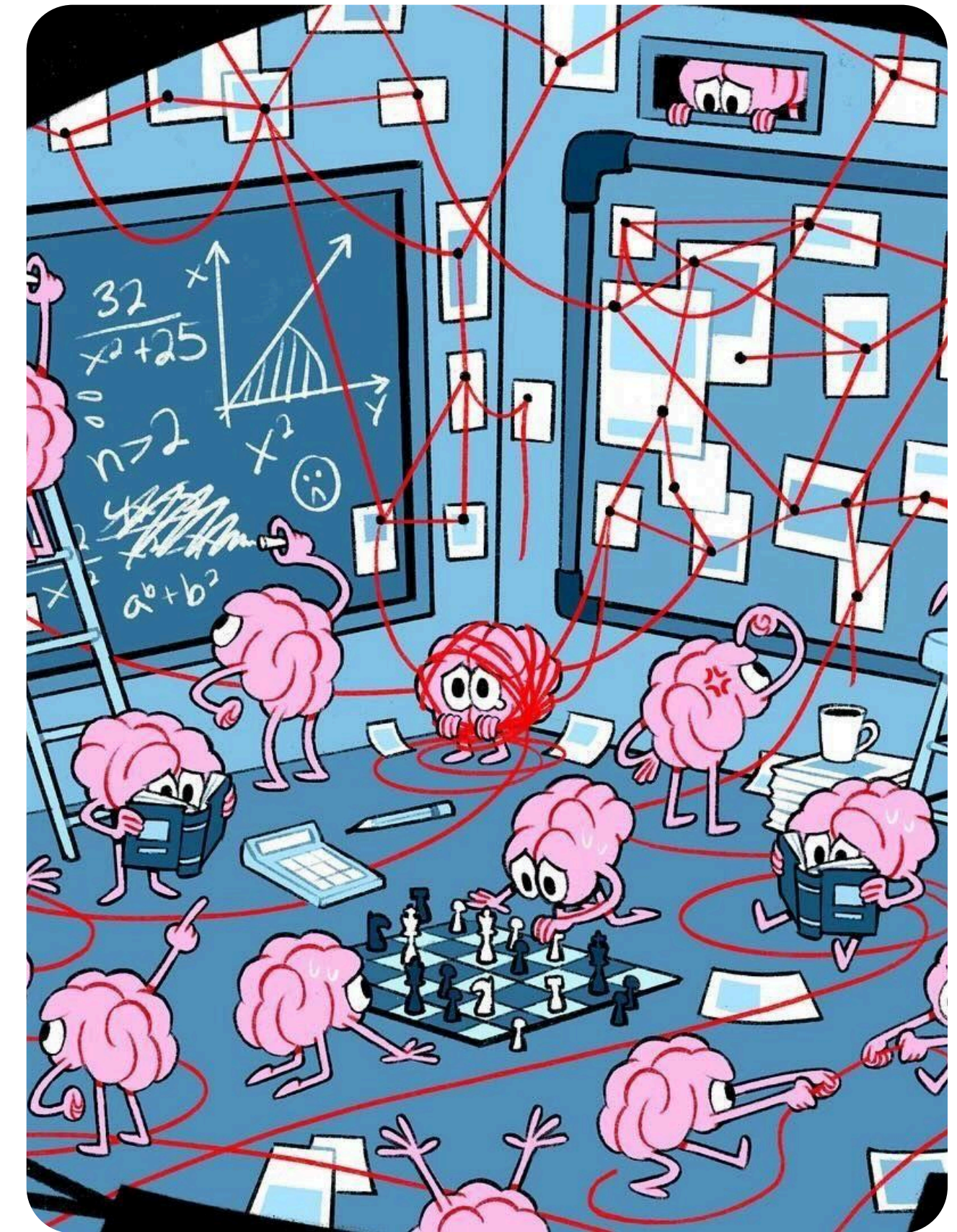
What's the main problems?



Memory system
overwhelmed by large data



Different platforms split up
knowledge and thinking



Saved content is hard to
find again

Background

What's the main problems?

We encounter too much fragmented information

The useful parts are scattered across different platforms

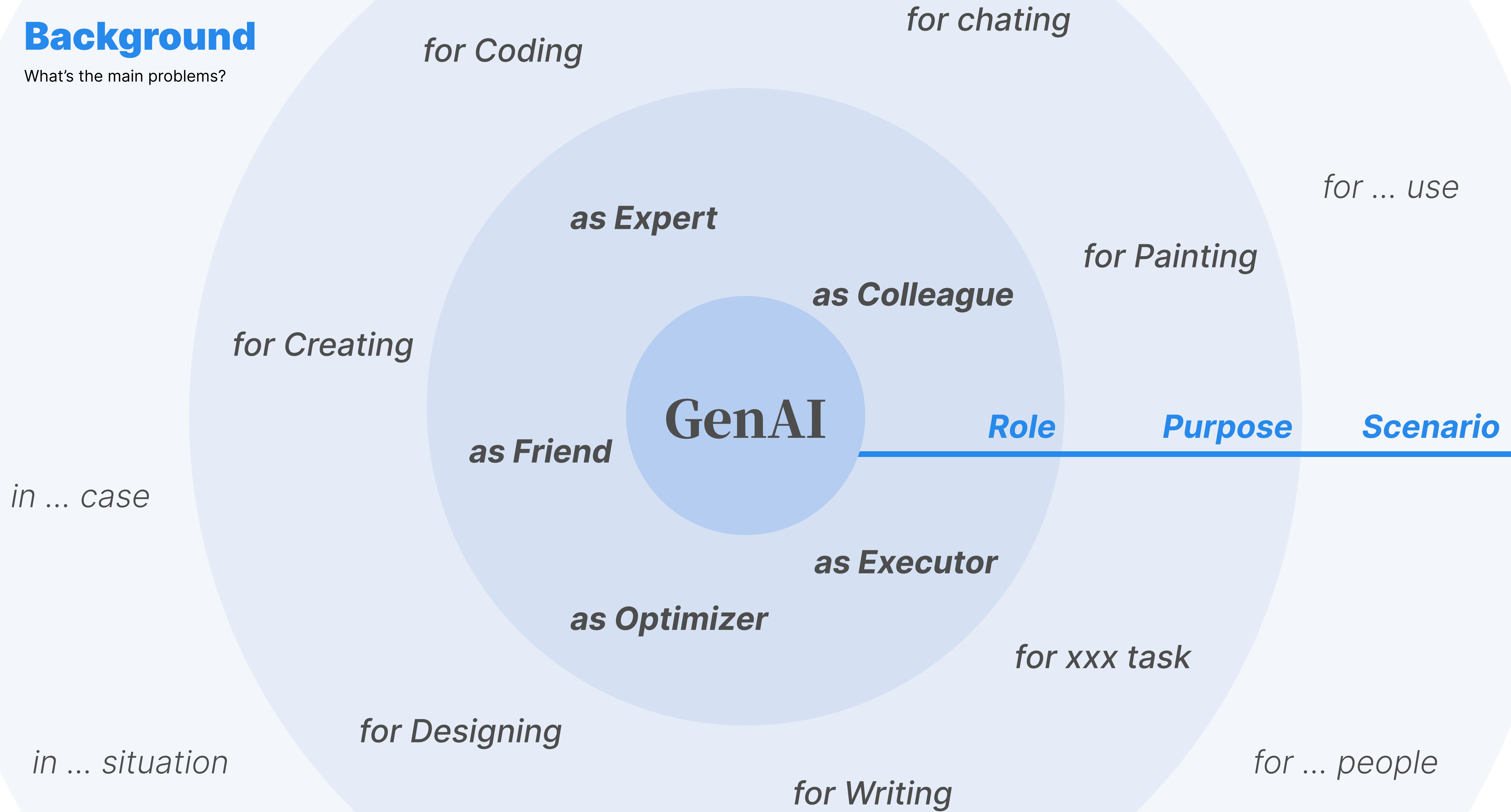
We save them somewhere, but it's hard to find them again.

“Cognitive poverty in the age of information abundance”



Background

What's the main problems?



Background

What's the main problems?

GenAI

as
Cognitive
Exoskeleton

Role

for externalizing,
structuring, and
activating knowledge

Purpose

in
cognitive
augmentation scenario

Scenario



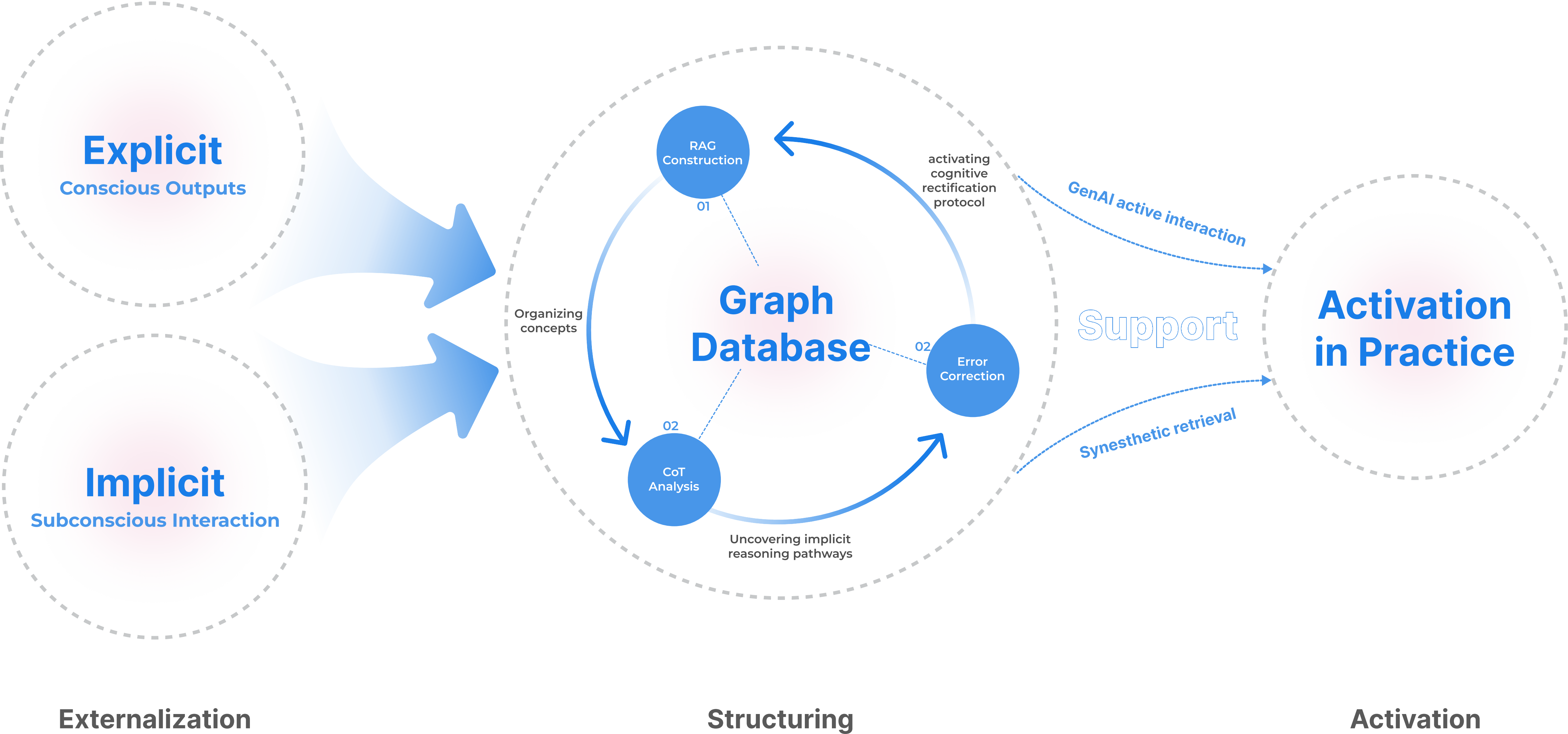
Brain

Generative AI as a Cognitive Exoskeleton for Externalizing, Structuring, and Activating Knowledge

Cache

Brain Cache

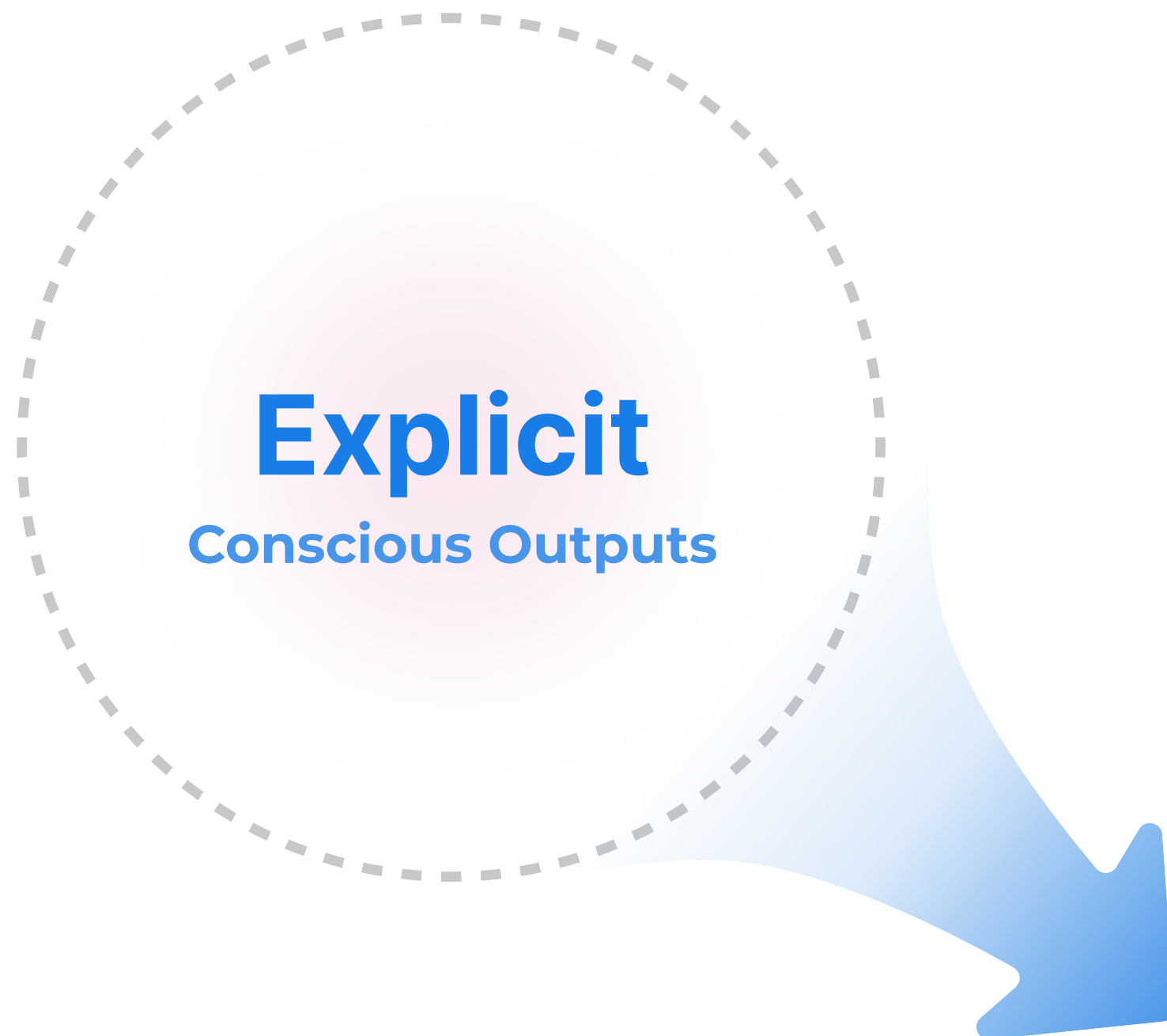
A Cognitive Augmentation Framework



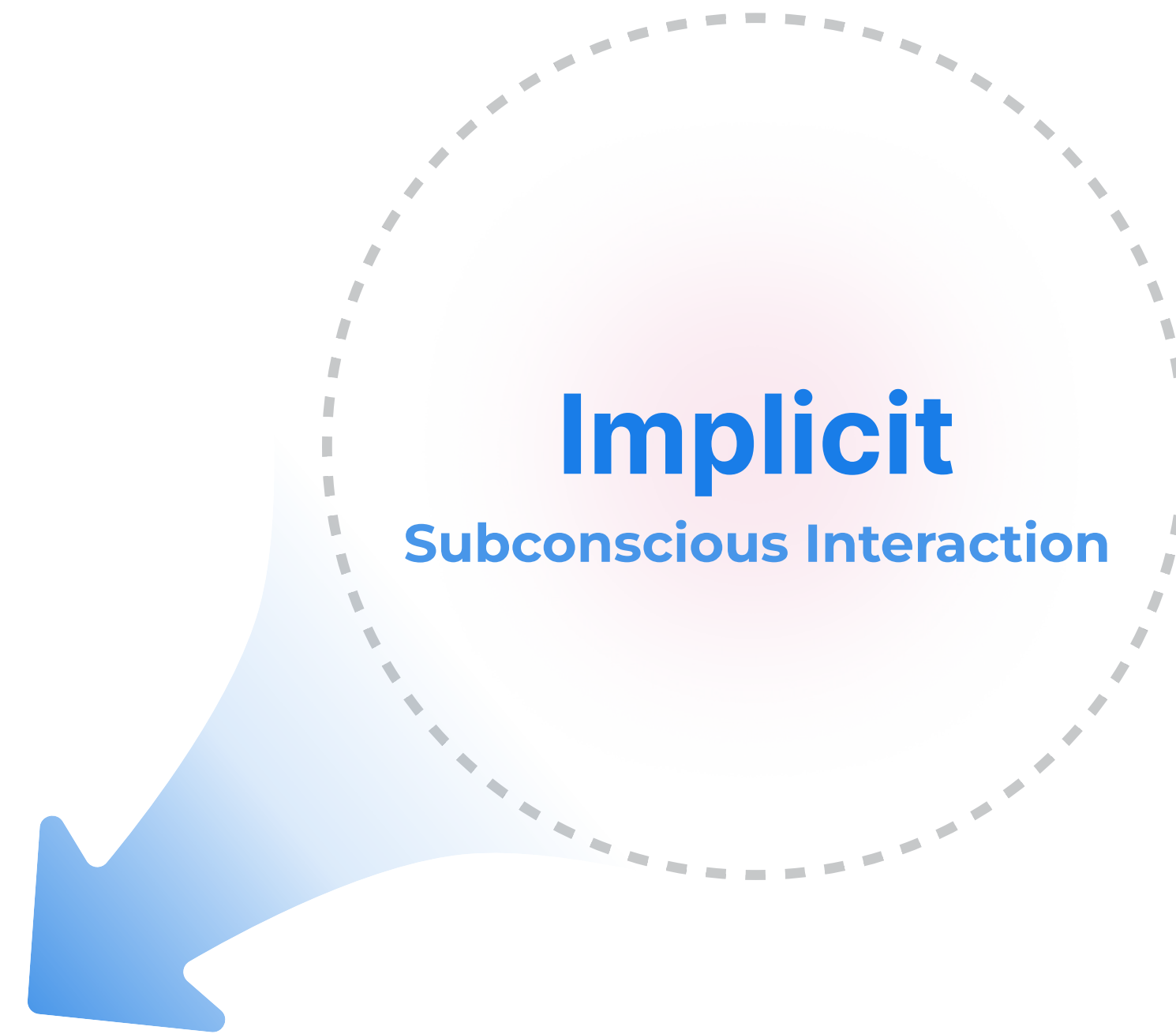
Externalization: Vectorized Memory Scaffolds

Transform the traces of thinking into external storage

Such as research notes, annotated diagrams, and lecture recordings with detailed contextual metadata



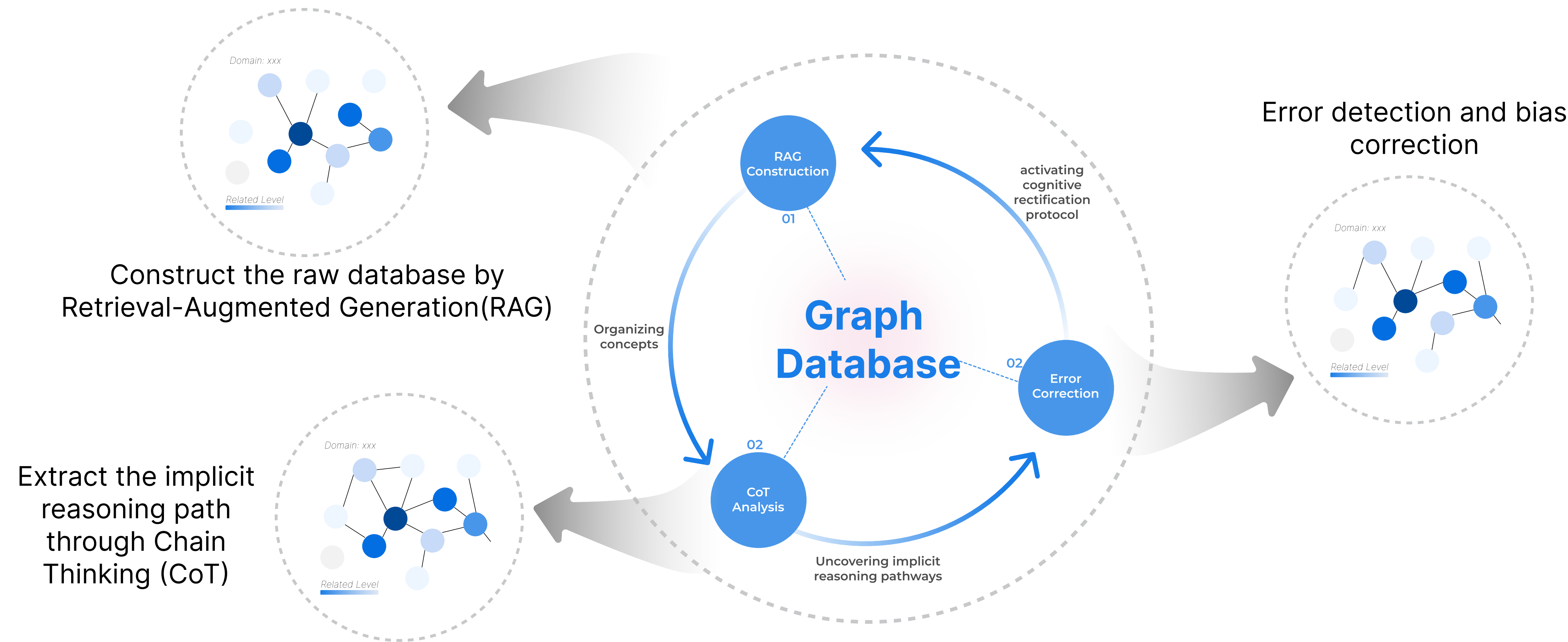
Such as interaction records, eye-tracking heatmaps, and EEG patterns



Merging both channels, the system creates a vector space that preserves the continuity of thought

Structuring: Iterative Cognitive Graphs

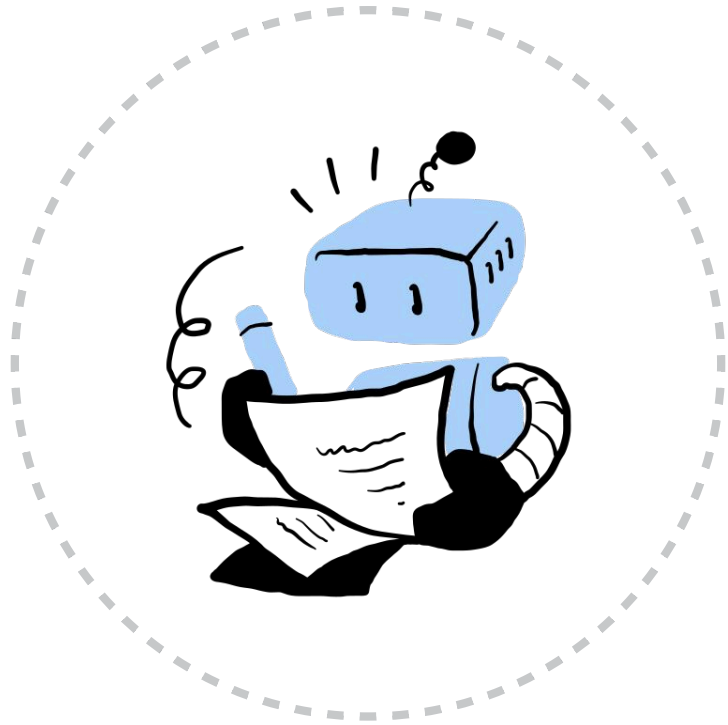
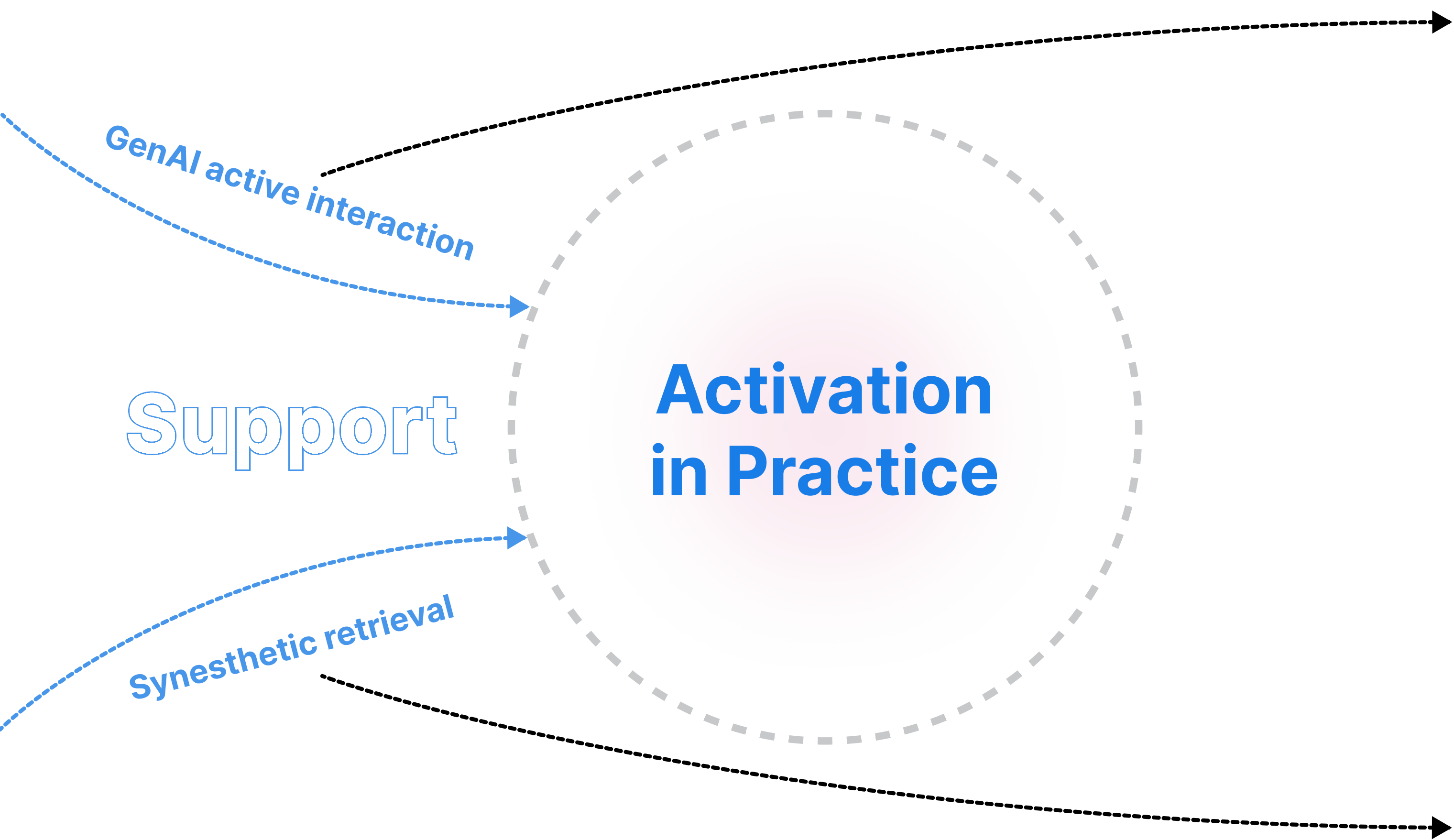
Organize the fragments into a cognitive map



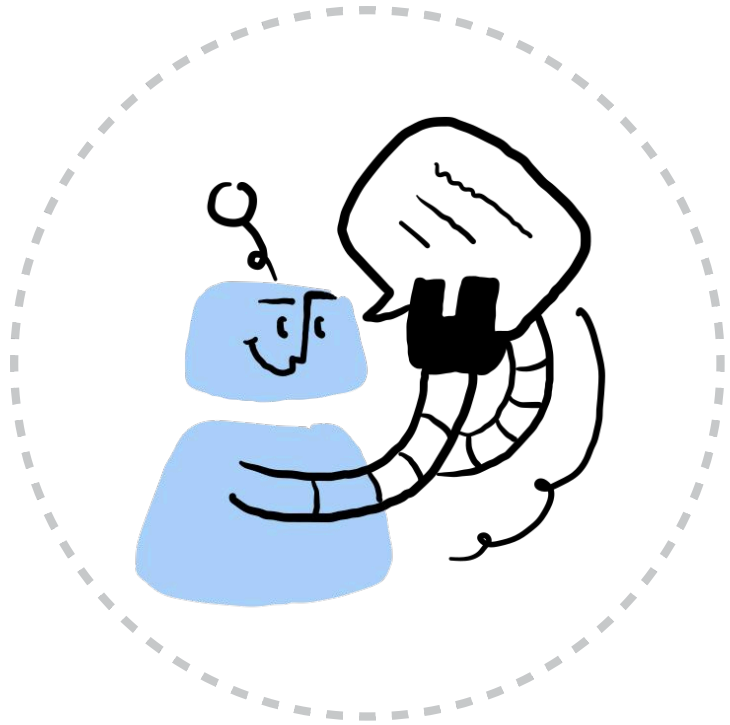
Dynamically evolving cognitive network: The originally chaotic nodes are organized into a causal chain

Activation: Associative Priming Engine

Organize the fragments into a cognitive map



Predict useful knowledge



Provide information proactively

▼ Title

Topic 1 is a demo version. We'll put the most important infos here. Topic 1 is a demo version. We'll put the most important infos here. Topic 1 is a demo version. We'll put the most important infos here.

▼ Topic 1-1

Topic 1 is a demo version. We'll put the most important infos here. Topic 1 is a demo version. We'll put the most important infos here. Topic 1 is a demo version. We'll put the most important infos here.

▼ Topic 1-2

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Tag:

Source: Guess you like

topic	topic	topic	topic

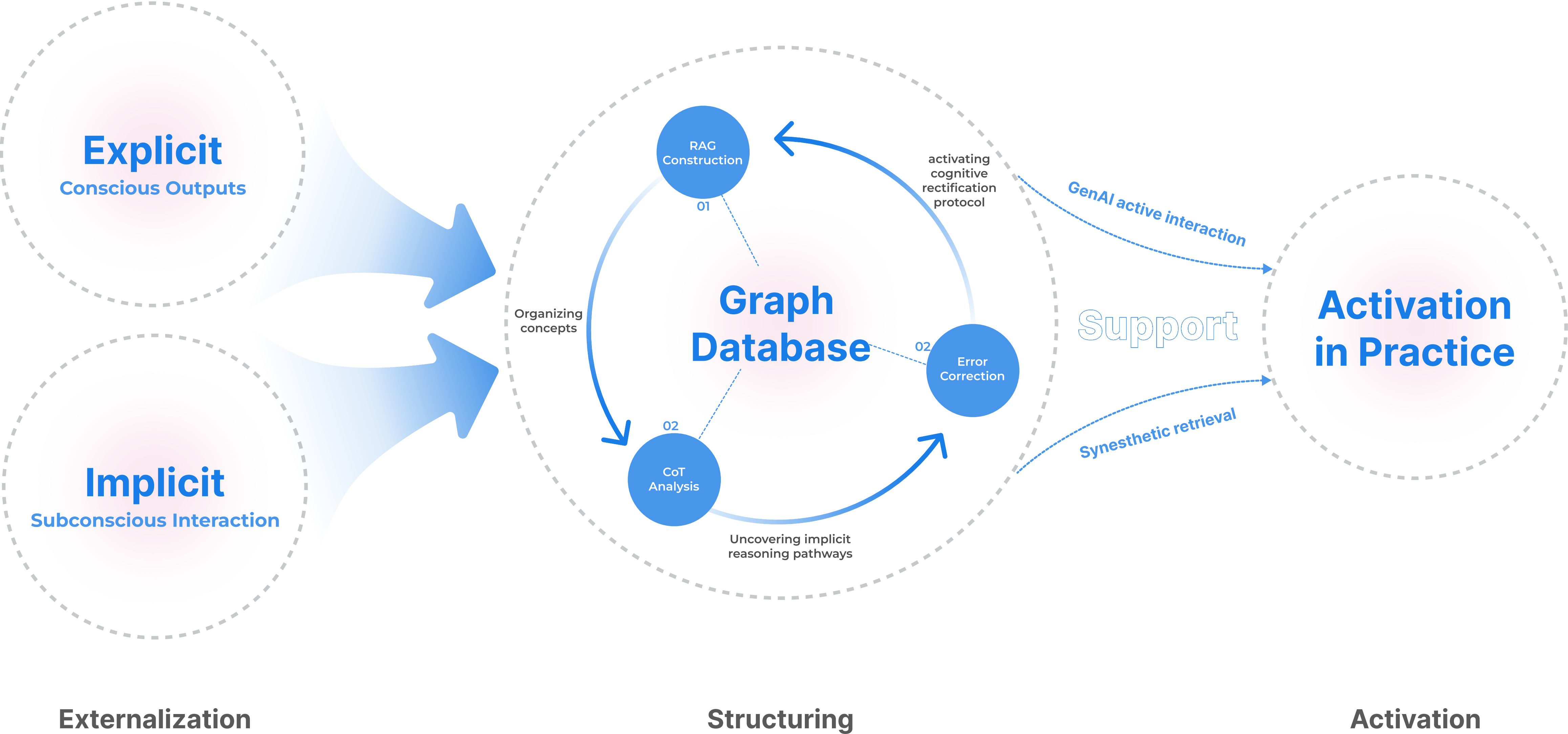
time time time time time time

Push content based on users' unique cognitive styles

Making the structured knowledge actionable and relevant in practice

Brain Cache

A Cognitive Augmentation Framework



Challenges and Reflections



Excessive reliance on external systems?

Is capturing implicit thinking legal and reasonable?

To whom does the externally generated knowledge belong?

...

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Thanks for Watching

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